months with re-Op, and 28.1 months with re-Op/RT. (p =0.066) Three risk factors (age>50, WHO grade IV, and unmethylated promoter of MGMT) were significantly associated with poor OS in multivariate analysis. Benefit of re-RT in both OS and PFS was established in patients carrying 2 or more risk factors. During re-RT, 4 patients (8%) had grade 2 or higher toxicity, and 3 patients (6%) did not complete re-RT. No radionecrosis was observed.

Conclusion: Re-RT after re-Op was tolerable with a cumulative median EQD_2 of 99.3 Gy and resulted in clear benefit in PFS and marginal gain in OS. Survival gain with re-Op/RT was more prominent in patients with two or more risk factors (age≥50, WHO pathologic grade IV, unmethylated MGMT promoter), and needs to be validated.

Author Disclosure: **S. Chun**: None. **I. Kim**: None. **S. Park**: None. **C. Park**: None. **J. Kim**: None. **T. Kim**: None. **S. Choi**: None. **S. Lee**: None.

MO_14_2592

TCP and Dose Response after Brachytherapy for Choroidal Melanoma

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Purpose/Objective(s): Brachytherapy is a commonly used eye-preserving treatment for choroidal melanomas. We hypothesized that the minimum tumor dose is related to risk of tumor recurrence.

Materials/Methods: We studied consecutive patients with primary choroidal melanomas, treated with Ru-106 plaque brachytherapy from 2005 to 2014 at a single institution to an apical dose of 100 Gy. Plans were retrospectively created using 3D image-guided planning software. Pre-treatment retinographies were used to contour the tumor; posttreatment retinographies to determine the accurate plaque position. Dose volume histograms were extracted. Patient and tumor characteristics, treatment details, and clinical outcomes were extracted from patient records. All patients were followed regularly until death or study cut-off (Jan 2018). Biologically effective doses (BED) were calculated using source half times, tissue specific factors ($\alpha/\beta=11.5$ Gy, $T_{1/2}=1.5$h), and a well-established model (Gagne et al, Med Phys, 2012) The relationship between tumor dose and risk of recurrence was examined using multivariate Cox regression modelling. Minimum physical dose to the tumor ($D_{99\%}$), transpupillary thermotherapy (TTT), tumor height, largest base dimension, incident eye, and gender were included in the model. Model robustness was assessed by considering alternative dose metrics, BED, and by taking the competing risk of death and TTT during follow-up into account (Aalen-Johansen estimator for cumulative incidence and Fine & Gray's model for regression analysis).

Results: We included 227 patients (108 females, 110 right eyes), with median treatment time of 120h (IQR: 74-191). Median tumor height and largest base dimension were 4 mm (IQR: 3-6) and 11 mm (IQR: 9-13). Median follow-up time was 4.9 years. The estimated 3-year local control was 80% (95% CI: 75-86). The median $D_{99\%}$ was 104 Gy (IQR: 63-138); this was the only significant factor associated with recurrence in multivariate modelling (p=0.0007). Hazard ratio (HR) for 10 Gy increase in $D_{99\%}$ was 0.90 (95% CI:0.84-0.96). Estimated 3-year local control for $D_{99\%}=50$ Gy and $D_{99\%}=100$ Gy were 70% (95% CI: 58-85) and 81% (95% CI: 73-91), respectively. The median $BED_{99\%}$ was 145 Gy (IQR: 73-196); using $BED_{99\%}$ in the multivariate model gave no substantial differences in estimates (HR 0.93 (95% CI: 0.90-0.97, p=0.001). Robustness checks with $D_{1-99\%}$ and $BED_{1-99\%}$ showed $D_{99\%}$ and $BED_{99\%}$ to be the most significant dose metrics for recurrence. Competing risk analysis showed cumulative incidences at 3 years for recurrence, death, and TTT of 17% (95% CI: 12-22), 14% (95 % CI: 9-18), and TTT 6% (95% CI: 3-9). Accounting for death and TTT, $D_{99\%}$ remained the only significant factor for recurrence (p=0.0004), with HR for 10 Gy increase in $D_{99\%}$ of 0.90 (95% CI: 0.85-0.95).

Conclusion: The minimum dose to the tumor correlated strongly with risk of tumor recurrence, with 100 Gy needed to ensure at least 80% local control at 3-years.

Author Disclosure: **C. Espensen**: Research Grant; Arvid Nilssons Foundation, Synoptik-Foundation, The Danish Eye Research Foundation (Øjenfonden), the Danish Cancer Research Foundation (Dansk Kræftforsknings Fond). **L. Fog**: None. **M.C. Aznar**: None. **A. Gothelf**: None. **J. Kiilgaard**: None. **A. Appelt**: None.

MO_14_2593

Long-Term Outcomes of Adult Patients with High-Risk Acute Lymphoblastic Leukemia Undergoing Total Body Irradiation with or without Whole Brain Boost

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Purpose/Objective(s): Total Body Irradiation (TBI) is an integral part of the conditioning regimen for patients with Acute Lymphoblastic Leukemia (ALL) undergoing allogeneic hematopoietic cell transplantation (allo-HCT). There is conflicting data in the literature regarding the utility of pre-transplant cranial irradiation in adult ALL patients without central nervous system involvement. The goal of this study is to investigate the post-transplant clinical outcomes of high-risk adult ALL patients undergoing allo-HCT with TBI conditioning with, or without, pre-transplant whole brain boost.

Materials/Methods: A retrospective cohort study was conducted utilizing medical record analysis. We identified 58 patients treated from January, 1998, to December, 2016, meeting our pre-set inclusion criteria of adults (age >18 years old) who carried a pathologically confirmed diagnosis of high-risk ALL (WBC >50,000 at diagnosis, T-cell ALL subtype, or high risk cytogenetic features), who underwent HCT with TBI conditioning. A total of 26 patient had undergone TBI alone, and 32 patients had undergone a cranial boost in addition to TBI. Additional data was collected including patient age, gender, TBI dose, Whole brain radiation dose, depth of response, previous SCT, WBC count at diagnosis, CNS relapse-free survival, progression-free survival, and overall survival. Multivariate analysis of correlation between categorical variables was assessed with Chi squared ($\chi 2$) analysis of independence. Survival analyses were assessed using Kaplan-Meier technique with a log rank test.

Results: Median follow up was 5.3 years. TBI dose consisted of 12Gy delivered in 6 fractions administered twice daily. Cranial boost consisted of an additional 6-14Gy administered to the whole brain. There was a statistically significant improvement in 5-year CNS relapse-free survival (100% vs. 77%; p=0.043) in favor of patients undergoing cranial boost. There was a trend towards a statistically significant improvement in 5-year progression-free survival (83.3% vs 65.4%; p=0.076) in favor of patients undergoing cranial boost. There was no difference in 5-year overall survival (50.0% vs 69.2%; p=0.921) with, versus without a cranial boost. On multivariate analysis, there was no statistically significant relationship between patient age, patient gender, TBI dose, cranial boost dose, depth of response to transplant, previous SCT, WBC count at diagnosis, and the survival endpoints. The presence of cranial boost was the only identified variable with an independent relationship to CNS relapse-free survival.

Conclusion: Adult patients with high-risk, CNS negative, ALL have a statistically significant improvement in CNS relapse-free survival, and may have an improvement in progression-free survival with the inclusion of a cranial boost with TBI pre-transplant conditioning. Our data suggest that the addition of a cranial boost may reduce the risk of CNS relapse in this high risk patient population.

Author Disclosure: **J.M. Famoso**: None. **J.L. Grow**: None. **B. Laughlin**: None. **E. Katsanis**: None. **B. Stea**: Stock Options; Salutaris MD.